BETHANY HIGH SCHOOL, Sarjapur, Bangalore
SELF ASSESSMENT
Std – VIII
CHEMISTRY

TOTAL MARKS: 30 TIME: 30 MINS

Question 1

Elements A, B and Z have atomic numbers 6, 9 and 12, respectively. Which one: [2]

a) Forms an anion

b) Forms a cation

Question 2

Metal M forms a chloride with the formula MCl_2. Write the formula of compound when M combines with: [3]

a) Sulphur

b) Nitrogen

c) Oxygen

Question 3 [3]

Draw the orbital diagrams of the following:

a) Magnesium chloride

b) Nitrogen

c) Hydrogen chloride

Question 4

An element X has 2 electrons in its M shell. It forms a bond with an element Y, which has 7 electrons in its third shell. [2]

a) Write the formula of the compound formed.

b) Which nearest inert-gas electronic configurations will elements X and Y acquire?

Question 5

Complete and balance the following chemical equations: [3]

a) $Mg + H_2SO_4 \rightarrow$

b) $Al + NaOH + H_2O \rightarrow$

c) $Zn + NaOH \rightarrow$

Question 6

Why are the following metals not used for the laboratory preparation of hydrogen? [3]

a) Calcium

b) Iron

c) Sodium

Question 7 [4]

a) Name two impurities present in hydrogen prepared in the laboratory.

b) How can these impurities be removed?

c) Write a balanced chemical equation for the reaction between water gas and excess steam? [mention all conditions]

d) How can hydrogen be purified from a mixture of hydrogen and carbon monoxide?

Question 8 [5]

State the type of reaction represented by each of the following equations and balance those that are not balanced.

a) $Br_2 + 2KI \rightarrow KBr + I_2$

b) $2HgO \rightarrow 2Hg + O_2$

c) $NH_4Cl \rightarrow NH_3 + HCl$

d) $PbO + 2HNO_3 \rightarrow Pb(NO_3)_2 + 2H_2O$

e) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

Question 9

(i) What do you observe when: [2]

a) Hydrated copper sulphate is heated.

b) An electric current is passed through acidified water.

(ii) Give one example of a reaction involving each of the following: [3]

a) Close contact

b) Pressure

c) Contact in solution

Answer key

Question 1

Elements A, B and Z have atomic numbers 6, 9 and 12, respectively. Which one: [2]

a) Forms an anion **[B and A]**

b) Forms a cation **Z**

Question 2

Metal M forms a chloride with the formula MCl_2. Write the formula of compound when M combines with: [3]

a) Sulphur **MS**

b) Nitrogen **M_3N_2**

c) Oxygen **MO**

Question 3 [3]

Draw the orbital diagrams of the following: **[Given in textbook]**

a) Magnesium chloride

b) Nitrogen

c) Hydrogen chloride

Question 4

An element X has 2 electrons in its M shell. It forms a bond with an element Y, which has 7 electrons in its third shell. [2]

a) Write the formula of the compound formed. **$MgCl_2$**

b) Which nearest inert-gas electronic configurations will elements X and Y acquire? **X= Neon, Y= Argon**

Question 5

Complete and balance the following chemical equations: [3]

a) $Mg + H_2SO_4 \rightarrow$ **$MgSO_4 + H_2\uparrow$**

b) $2Al + 2NaOH + 2H_2O \longrightarrow$ **$2NaAlO_2 + 3H_2\uparrow$**

c) $Zn + 2NaOH \longrightarrow$ **$Na_2ZnO_2 + H_2\uparrow$**

Question 6

Why are the following metals not used for the laboratory preparation of hydrogen? [3]

a) Calcium **[Expensive and vigorous reaction]**

b) Iron **[Reversible reaction]**

c) Sodium **[reacts very violently with water]**

Question 7 [4]

a) Name two impurities present in hydrogen prepared in the laboratory.

b) How can these impurities be removed?

c) Write a balanced chemical equation for the reaction between water gas and excess steam? [mention all conditions] **$CO + H_2 + H_2O \rightarrow CO_2 + H_2$ Conditions: Excess steam, Temperature 450^0C, catalyst Fe_2O_3**

d) How can hydrogen be purified from a mixture of hydrogen and carbon monoxide? **The mixture is passed through ammoniacal cuprous chloride (CuCl) solution.**

Question 8 [5]

State the type of reaction represented by each of the following equations and balance those that are not balanced.

a) $Br_2 + 2KI \rightarrow 2KBr + I_2 \rightarrow$ **Synthesis**

b) $2HgO \rightarrow 2Hg + O_2 \rightarrow$ **Thermal Decomposition**

c) $NH_4Cl \rightarrow NH_3 + HCl \rightarrow$ **Thermal Decomposition**

d) $PbO + 2HNO_3 \rightarrow Pb(NO_3)_2 + 2H_2O \rightarrow$ **Neutralization**

e) $AgNO_3 + NaCl \rightarrow AgCl\downarrow + NaNO_3 \rightarrow$ **Principate reaction**

Question 9

(i) What do you observe when: [2]

a) Hydrated copper sulphate is heated. **Blue hydrated copper sulphate changes to white anhydrous copper sulphate**

b) An electric current is passed through acidified water. **Hydrogen is produced at the cathode, Oxygen is produced at the anode.**

(ii) Give one example of a reaction involving each of the following: [3]

a) Close contact **$4P + 5O_2 \rightarrow 2P_2O_5$**

b) Pressure **$N_2 + 3H_2 \rightleftharpoons 2NH_3$ [200 atms]**

c) Contact in solution **$NaOH + HCl \rightarrow NaCl + H_2O$**